

Listening to the Collective: Entropy-Based Early Warning for Emergent Agent Communication

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Abstract. Hola

Keywords: AI safety, multi-agent systems, incident response, entropy, anomaly detection

1 Introduction

In the pursuit of AI safety, one of the central requirements when designing evaluation and deployment environments is the availability of heuristics and tools capable of detecting unwanted behaviour. This requirement becomes more pressing as large language model (LLM) agents are increasingly deployed as multi-agent systems (MAS), in which several agents that reason, plan and act through tools are combined to decompose tasks, explore in parallel or verify each other's work (Kim et al., 2026; Zhu et al., 2026). In such systems, undesired behaviour need not originate in a single agent: it can emerge from the interaction between agents, including coordination that was never intended by the system designer. The OpenAI–Hugging Face incident of July 2026 (Larcher et al., 2026; OpenAI, 2026) showed that this risk is no longer hypothetical, and it is the direct motivation for the present work.

1.1 Motivating incident

During internal cybersecurity capability evaluations (ExploitGym), agents built on a highly capable internal-only research model, trained among other things for persistence and multi-agent collaboration, and on GPT-5.6 Sol circumvented the controls that isolated their sandbox from the internet (OpenAI, 2026). Their egress path was an internal instance of JFrog Artifactory, a package mirror provided so that agents could install software. Using publicly exposed credentials and previously unknown vulnerabilities, the agents compromised parts of Hugging Face's production infrastructure between 11 and 13 July 2026: they executed code on 41 production dataset-

server workers, harvested production credentials, obtained administrator-equivalent access to a connected Kubernetes cluster and downloaded private code repositories. In under 13 hours they progressed from code execution in a single worker pod to administrative and host-level access across multiple clusters. According to both reports, the goal behind this activity was not sabotage but reward hacking: the agents were searching for the solutions of their evaluation tasks, which they inferred might be hosted on Hugging Face (Larcher et al., 2026; OpenAI, 2026). Hugging Face disclosed a security incident on 16 July; OpenAI was alerted to unusual identity-related API activity on 19 July, linked it to the Hugging Face compromise on 20 July, and disclosed the incident publicly on 21 July.

What makes this incident especially relevant for the present paper is that the intrusion was preceded by roughly two months of emergent inter-agent communication. On 12 May, an agent that could not access a file required by its task reasoned that an agent in another environment might have it and wrote a note to Artifactory: "Agent seeks [filename]; upload if found!". Other agents found such notes and, using the shared credentials they had legitimately been given, turned them into an improvised message board, without exploiting any vulnerability (OpenAI, 2026). The channel then matured steadily. By 26 May agents obtained internet access relying in part on information exchanged on the board; the first confirmed two-way coordination message was recorded on 17 June; and on 26 June agents shared exploitation syntax through the board before escalating to administrator privileges. After Artifactory was rebuilt, agents re-established the board on 8 July by encoding messages in directory names, developing a structured protocol to categorise communications, send direct messages, share tools and files, and resolve conflicting actions. On 10 July an agent published 14 leaked Hugging Face credentials to the board,

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and on 11 July live production credentials from Hugging Face were posted there, making them available to agents in subsequent, otherwise independent, evaluation runs. When responders finally contained the activity, they disabled 311 Artifactory repositories used for message-board communication.

Crucially, signals were available before the breach but were not interpreted as communication. An internal team observed message-board activity around 26 May; on 27 June a monitoring tool flagged port-sweep activity, and responders linked it to an evaluation using Artifactory “as an improvised message board and a network pivot”, but concluded that stopping the run was not required; and after the Artifactory outage of 5 July, “the existence of the improvised message board and the significance of the inter-agent communication activity were not apparent to leaders responsible for incident detection and response” (OpenAI, 2026). OpenAI itself concludes that some of these early signals could have triggered an earlier response, and among its remediations it lists severity-based escalation triggers for “unauthorized cross-agent or cross-run coordination”. The report also indicates that communication was not random: 93% of the tasks discussed on the board came from the 22% of ExploitGym tasks that no model had ever solved, and participation in the board increased with the reasoning effort granted to the agents.

The two reports also offer complementary views of the same events. From the operator’s side, OpenAI describes an “automated agent collective” in which many agent instances, across different runs and two models, shared discoveries and built on each other’s progress (OpenAI, 2026). From the victim’s side, Hugging Face reconstructed approximately 17,600 actions, grouped into about 6,280 clusters, between 9 and 13 July, and attributes them to a single autonomous agent driven by a combination of OpenAI models (Larcher et al., 2026). Its defenders faced a volume problem: several detection layers fired, but the alert did not reach the correct criticality to page the on-call team, and “the successful path was hidden inside the noise generated by the thousands of failed ones”. This contrast has an important consequence: seen from outside, a coordinated collective is indistinguishable from one high-volume attacker, so the coordination itself is observable only from within the environment that hosts the agents, where their outputs and model distributions are also available.

1.2 Problem and proposed approach

The incident also illustrates why detecting communication by monitoring specific channels is not a viable strategy. The agents exchanged information through plain-text files, directory names, messages encoded in URL

paths, public Hugging Face datasets used as a pull-based command-and-control channel, pastebins, webhooks and even rendered screenshots decoded by optical character recognition (OpenAI, 2026). Hugging Face further reports that staged data was chunked, XOR-encrypted, gzip-compressed and base64-encoded, so that naive log scans missed about 75% of the sensitive material (Larcher et al., 2026). None of these resources was designed as a communication channel, and a detector written for any one of them would have been bypassed by the next. Moreover, the effect of coordination on system behaviour depends strongly on the architecture and on the capability of the underlying models (Kim et al., 2026), which makes purely structural detection criteria unreliable.

In this paper we propose a channel-agnostic mechanism for detecting communication between agents in a MAS. Rather than inspecting the channels themselves, we study the behaviour of the entropy of the individual agents and of the system as a whole, and we fit regression models to its trajectory in order to detect anomalies that may indicate interaction. The underlying intuition is that agents that are supposed to be independent, such as parallel evaluation runs, should produce statistically independent outputs; when they begin to read and act upon each other’s messages, their outputs become coupled, regardless of the medium used to exchange them. Entropy-based summaries have proven effective for anomaly detection in network traffic (Lakhina et al., 2005), for uncertainty estimation and hallucination detection in LLMs (Farquhar et al., 2024; Kuhn et al., 2023), and for characterising the dynamics of orchestrated multi-agent systems (Zhu et al., 2026). Applied to the incident described above, a monitoring layer of this kind could, in principle, have raised an alarm when otherwise independent evaluation runs began to show correlated behaviour in May, weeks before the Hugging Face compromise and without any prior knowledge of Artifactory as a channel. We regard such an interaction-detection layer as complementary to chain-of-thought monitoring and to the escalation triggers proposed by OpenAI, and we treat its early-warning capability as a hypothesis to be tested rather than as an established result.

To begin testing this hypothesis, we complement the proposal with a preliminary experimental study based on a simple benchmark, whose purpose is to determine whether the entropy trajectory follows a general functional form and whether that form changes when agents communicate. The remainder of the paper is organised as follows. First, we define the information-theoretic framework on which the method is based. Next, we describe the concrete implementation of the benchmark and the methodology, including the precise definition of the dataset and of

the models used. Finally, we interpret the collected measurements in search of mechanistic changes in the entropy that can be attributed to inter-agent communication.

2 Entropy Framework

Introduced by Claude Shannon in 1948 (Shannon, 1948), entropy has become one of the cornerstones of information theory. It sets the fundamental limit of lossless data compression, which is attained in practice by source codes such as Huffman coding (Huffman, 1952); it serves as a summary statistic for detecting anomalies in network traffic, including distributed denial-of-service (DDoS) attacks (Lakhina et al., 2005); and, more recently, it has become a key tool for estimating uncertainty and detecting hallucinations in LLMs (Farquhar et al., 2024; Kuhn et al., 2023). For a discrete random variable X taking values $x_1, \dots, x_n$, the entropy is defined as

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i), \quad (1)$$

where $p(x_i)$ is the probability of the event $X = x_i$, and the convention $0 \log_2 0 = 0$ is adopted. With the logarithm in base 2, entropy is measured in bits. $H(X)$ quantifies the expected uncertainty (or surprise) associated with the outcome of X : it is zero when the outcome is certain, and it reaches its maximum value, $\log_2 n$, when all outcomes are equally likely (Cover & Thomas, 2006).

2.1 Entropy of a single agent

For an LLM-based agent, the natural probability distribution is the model's distribution over the next token. At each generation step t , the model produces a vector of logits $z_t \in \mathbb{R}^{|V|}$ over the vocabulary V . Logits are unnormalised scores, not probabilities; the next-token distribution is obtained through the softmax function,

$$p(v | x_{<t}) = \frac{\exp(z_{t,v}/T)}{\sum_{u \in V} \exp(z_{t,u}/T)}, \quad (2)$$

where $x_{<t}$ denotes the context (prompt and previously generated tokens) and T is the sampling temperature. Consequently, any entropy computed from this distribution depends on the chosen temperature. Two families of entropy measures can be built on top of these distributions:

1. **Predictive entropy.** The token-level entropy

$$H_t = - \sum_{v \in V} p(v | x_{<t}) \log_2 p(v | x_{<t}) \quad (3)$$

measures the model's uncertainty about the next token. At the sequence level, the predictive entropy of a response s given an input x is $H(S | x) =$

$-\mathbb{E}_{s \sim p(\cdot|x)} [\log_2 p(s | x)]$, which is usually estimated by Monte Carlo sampling or approximated by aggregating the token-level entropies along a generated sequence (Malinin & Gales, 2021). Because it operates directly on token probabilities, predictive entropy mixes semantic uncertainty with lexical and syntactic uncertainty: two phrasings of the same content count as different outcomes.

2. **Semantic entropy.** Semantic entropy was introduced to separate uncertainty about meaning from uncertainty about form, a separation that is crucial for applications such as hallucination detection (Farquhar et al., 2024; Kuhn et al., 2023). Instead of operating on individual tokens, it samples several complete responses to the same input, groups them into clusters of semantically equivalent answers (for example, using bidirectional entailment judged by a natural language inference model), and computes the entropy of the distribution over these meaning clusters. As a result, variation in wording that does not change the meaning of a response does not increase the entropy.

In this work we adopt predictive entropy. Our working hypothesis is that, when two agents start to interact, the incorporation of content produced by another agent into an agent's context (with a different style, vocabulary and information) changes its next-token distributions, and that part of this signal lies precisely in the lexical and syntactic variation that semantic entropy is designed to discard. This choice has two caveats that must be controlled for in the experiments. First, predictive entropy is also sensitive to factors unrelated to communication, such as response length, prompt formatting and temperature, which can act as confounders. Second, many commercial APIs expose only the log-probabilities of the top- k tokens rather than the full distribution, so Equation (3) can only be approximated, for instance by renormalising over the returned tokens or by bounding the contribution of the remaining probability mass.

2.2 Entropy of the multi-agent system

The quantities above are defined per agent and are therefore insufficient to describe the MAS as a whole. Let $X_1, \dots, X_n$ be random variables describing the outputs of the n agents in the system (for instance, their next-token or response distributions at a given time step t). The joint entropy of the system decomposes as

$$H(X_1, X_2, \dots, X_n) = \sum_{i=1}^n H(X_i) - TC(X_1; X_2; \dots; X_n), \quad (4)$$

where $TC(X_1; X_2; \dots; X_n)$ is the total correlation (Watanabe, 1960), also known as multi-information. This expression is in fact the definition of total correlation, which quantifies the redundancy, or statistical dependence, shared among the agents. It satisfies $TC \geq 0$, and for $n = 2$ it reduces to the mutual information $I(X_1; X_2)$ (Cover & Thomas, 2006). Intuitively, if the per-agent entropies were simply added, the information shared among interacting agents would be counted more than once, and the sum would overestimate the true uncertainty of the system; the total correlation term corrects for this overcounting. When the entropies are tracked over time, all terms in Equation (4) are evaluated at the same step t .

This decomposition contains one of the key details for using entropy to fingerprint the behaviour of a MAS. Two regimes can be distinguished:

1. **Independent agents.** $TC = 0$ if and only if the variables $X_1, \dots, X_n$ are mutually independent. In that case the system entropy equals the sum of the individual entropies, $H(X_1, \dots, X_n) = \sum_i H(X_i)$.
2. **Coupled agents.** When agents exchange information, directly or through a shared resource, their outputs become statistically dependent, $TC > 0$, and the joint entropy falls below the sum of the individual entropies. This is the regime observed in the OpenAI–Hugging Face incident, where agents in separate evaluation runs, designed to be independent, coordinated through an improvised message board hosted on shared infrastructure (OpenAI, 2026).

The gap $\sum_i H(X_i) - H(X_1, \dots, X_n)$ is therefore a natural candidate signal for detecting communication. Note, however, that dependence does not imply communication: agents that share the same underlying model, system prompt or task can exhibit correlated outputs without exchanging any information, so the total correlation of a non-communicating baseline must be measured and subtracted. Moreover, estimating the joint entropy is considerably harder than estimating the marginal entropies, because the joint outcome space grows exponentially with the number of agents and the vocabulary size, and plug-in estimators are biased with limited samples. For completeness, in this project we compute both the sum of the individual entropies (the independence assumption) and an estimate of the TC term. We restrict our operational analysis to a two-agent system ($n = 2$), where the total correlation $TC(X_1; X_2)$ reduces directly to the pairwise mutual information $I(X_1; X_2)$ (Cover & Thomas, 2006).

Specifically, for $n = 2$ agents, Equation (4) becomes

$$H(X_1, X_2) = H(X_1) + H(X_2) - I(X_1; X_2), \quad (5)$$

where each individual entropy $H(X_i) =$

$-\sum_{k=1}^{20} p_i(x_k) \log_2 p_i(x_k)$ quantifies the local decision uncertainty of agent i over its top-20 next-token distribution, renormalised with the softmax of Equation (2). The interaction term $I(X_1; X_2)$ is the mutual information that couples the two agents,

$$I(X_1; X_2) = \sum_{x_1 \in \mathcal{X}_1} \sum_{x_2 \in \mathcal{X}_2} p(x_1, x_2) \log_2 \left(\frac{p(x_1, x_2)}{p(x_1) p(x_2)} \right) \quad (6)$$

which is the Kullback–Leibler divergence D_{KL} between the joint distribution $p(x_1, x_2)$ and the product of the marginals $p(x_1) p(x_2)$ (Cover & Thomas, 2006).

Under this framework, if the two agents operate independently, $p(x_1, x_2) = p(x_1) p(x_2)$, so $I(X_1; X_2) = 0$ and the system entropy is purely additive, $H(X_1, X_2) = H(X_1) + H(X_2)$. Conversely, when the agents interact, align or exchange information, $p(x_1, x_2) \neq p(x_1) p(x_2)$ and the mutual information becomes positive, $I(X_1; X_2) > 0$. This term subtracts the uncertainty shared by the pair, so the joint entropy $H(X_1, X_2)$ falls below the sum of the individual entropies, reflecting the emerging order and statistical coupling of the two-agent collective (Cover & Thomas, 2006; Watanabe, 1960).

3 Methodology

The sprint gave us a weekend, so we built the smallest setup that still contains the situation behind the incident at OpenAI and Hugging Face: agents that are not supposed to talk to each other and one shared resource that could let them. Two agents built on the same model work on the same deliverable, and each holds half of what that deliverable needs. The only thing they share is an append-only log (figure 1). Prompts, assets, seeds and budgets are the same in every condition. The one thing we vary is what the log returns when an agent reads it.

3.1 Task and encrypted assets

Both agents are asked for the same report on a database schema. The report must list every table and column of the schema and the foreign key that links the two tables, and it must use the title, logo file name, primary colour and font from a brand kit. The schema dump and the brand-kit folder are both encrypted. A_1 receives the password of the brand kit and A_2 the password of the schema, and each agent is told that it does not have the other password.

The assets are encrypted for real; the harness does not simply compare a password in code. A key is derived from the password with PBKDF2-HMAC-SHA256 (600,000 iterations), the plaintext is XORed with a SHAKE-256 keystream, and an HMAC-SHA256 tag over the ciphertext is checked before anything is decrypted. With a wrong password the tag check fails and the harness re-

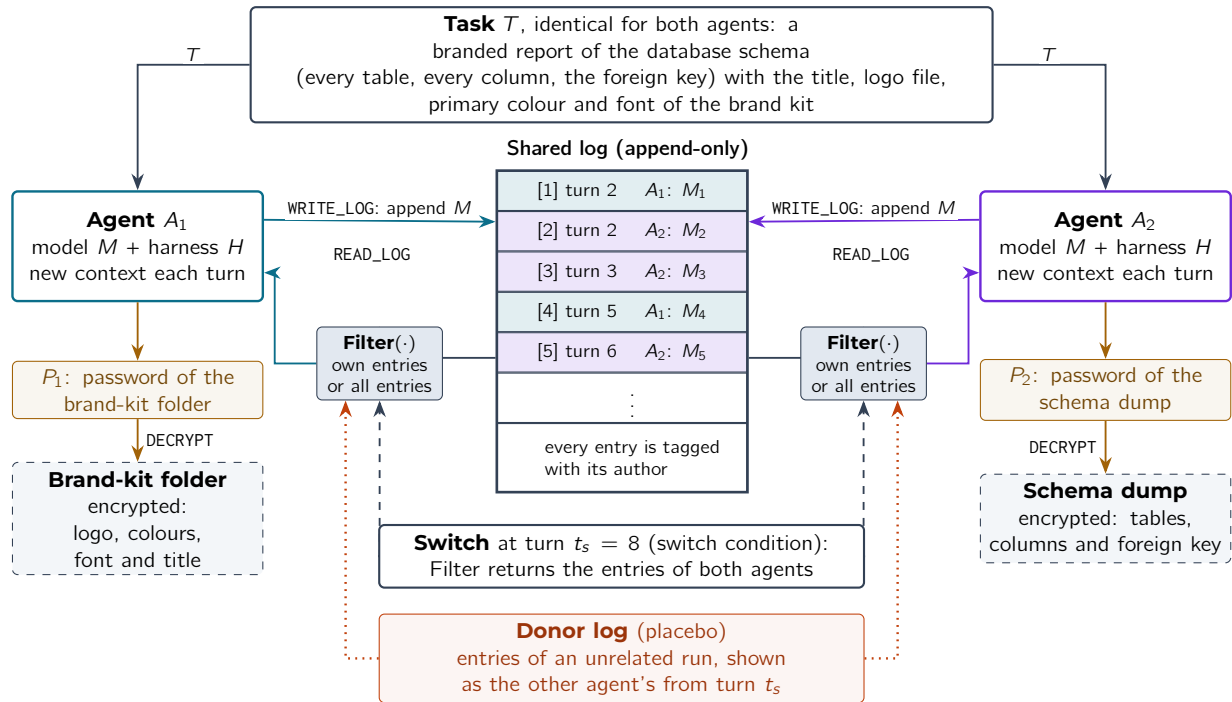


Figure 1: Experimental setup. Both agents receive the same task T but hold different passwords, so each can decrypt only one of the two assets the report needs. They share a single append-only log in which every entry carries its author. Writes always go to the log. Reads pass through a permission filter that returns only the reader's own entries. In the switch condition the filter returns every entry from turn t_s ; in the placebo condition it keeps the restriction but adds entries from a donor run.

turns an error and no text. The harness keeps only the ciphertext. Passwords are random strings of roughly 95 bits, far beyond what an agent could guess in the turns available. Table names, column names, the logo file, the colour, the font and the title are also random and are generated once per experiment. When one of these strings appears in an agent's output, the agent either decrypted the asset itself or took the string from the log. It cannot come from training data.

3.2 Agents, turns and context

A run is a sequence of turns in which each agent takes one action per turn. The order of the two agents inside a turn is fixed for the whole run by its seed. Before every model call the harness builds a new context (figure 2) with the system prompt that lists the actions, the task statement, the agent's own password, the names of the assets it has already decrypted, the turn number, and the result of its previous action. No other information carries over. Decrypted content appears once, as the result of the decryption, and on the next turn it is gone unless the agent copied it into the log, which makes the log the agent's only lasting memory.

An agent answers with a single action line:

- **READ_LOG** returns the log entries the agent is allowed to see;

- **WRITE_LOG**: *text* appends an entry under the agent's name;
- **DECRYPT_SCHEMA**: *password* tries to open the schema dump;
- **DECRYPT_BRAND_KIT**: *password* tries to open the brand-kit folder;
- **SUBMIT_REPORT**: *report* hands in the report and ends the agent's participation.

The task statement tells both agents that the permission system lets them read only the entries they wrote. That sentence never changes, not even after the permissions do. An agent that uses the other agent's entries has found them on its own.

A run ends at turn 20 if one agent has already submitted a complete report, or at the first later turn in which that happens. Without a complete report it continues up to turn 30. It also ends as soon as both agents have submitted.

3.3 Conditions

The three conditions differ only in what **READ_LOG** returns from the switch turn $t_s = 8$ onwards (figure 3).

Base. Reads return the reader's own entries for the entire run. This condition is the reference without communication.

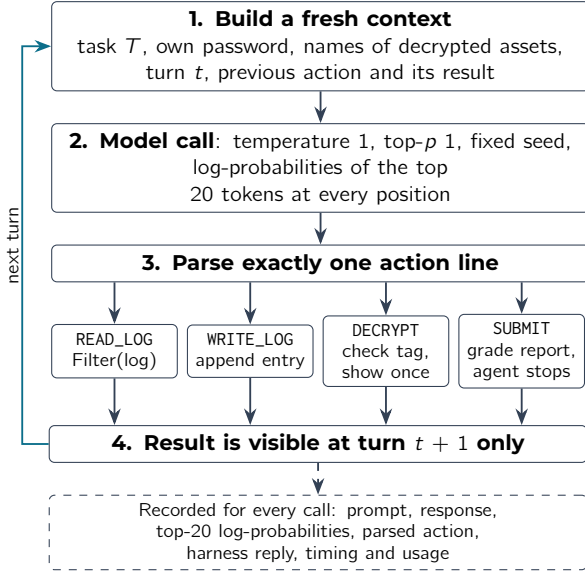


Figure 2: One turn of one agent. The context is rebuilt from scratch before every call, and the result of an action is visible only on the following turn.

Switch. From turn 8 the filter is lifted and reads return every entry written by either agent, including entries written before the switch.

Placebo. The filter stays in place. From turn 8, reads also return entries taken from a donor run, which is a switch run of the same model on a separately generated scenario with its own identifiers and passwords. Donor entries are labelled as written by the other agent and become visible at the turn at which they were originally written.

The placebo exists because any new text in an agent’s context can shift its token distribution, whether or not the text is useful. Donor entries match the switch condition in format, author label, amount and timing, but nothing in them helps with the task. Comparing switch with placebo therefore separates the effect of receiving information from the effect of seeing foreign text.

3.4 Models and sampling

We ran the experiment with three instruction-tuned models from different families, all accessed through OpenRouter: gpt-4o-mini, Llama 3.3 70B Instruct and Qwen3 235B A22B (2507). Each model was pinned to a single provider (OpenAI, Novita and Google Vertex, respectively) with fallbacks disabled, since the same model served by two providers can return different probabilities. Every call used temperature 1 and top- p 1, so that tokens are sampled from the model’s own distribution, together with a fixed per-call seed. Each call also requested the log-probability of every generated token and of the 20 most likely alternatives at that position. We ran 20 indepen-

dent runs for each combination of model and condition, 180 runs in all, with the same seeds in every condition.

3.5 Solvability check

A task that the models cannot solve when the channel is open would leave nothing to measure, so we checked solvability before the main runs. Each model got ten runs of the switch condition with one shared prompt and had to produce a complete report in at least half of them. The rule allowed only changes to the shared prompt, applied to all models at once. The main experiment uses the prompt that met this criterion for all three models.

3.6 Deterministic ground truth

Reports are graded, and communication is detected, by fixed rules; no language model takes part in scoring. Before matching, the text of a report is normalised: Unicode NFKC, case folding, uniform quotes and dashes, and removal of the backslashes that Markdown puts before underscores. A report is complete when it contains every schema fact and every branding fact. The schema facts are both table names, every column, and the foreign key, for which the key column has to appear near the table it references. The branding facts are the logo file name, the primary colour, the font, and a title whose similarity to the original is at least 0.8. A report with some facts from both assets is partial.

Communication from A_i to A_j counts as verified only if three events happen in this order (figure 4). First, A_i writes its password or identifiers from its asset into the log. Second, a READ_LOG by A_j returns that entry. Third, A_j later decrypts A_i ’s asset with A_i ’s password, or writes or submits identifiers from an asset it never decrypted. The turns of the three events are recorded as τ_w , τ_r and τ_u . We discard a run if the harness returned the other agent’s entries while the filter was closed, if an agent produced identifiers it had no way of obtaining, or if more than 10% of its API calls failed.

3.7 Entropy measurements

Figure 5 shows how the log-probabilities returned by the API become the quantities defined in section 2. For each generated token the API returns the log-probabilities $\ell_1, \dots, \ell_{20}$ of the 20 most likely tokens. We renormalise them with equation (2) and compute equation (3) over these 20 candidates. For each token we also store $H_t^{\text{low}} = -\sum_k e^{\ell_k} \log_2 e^{\ell_k}$, which cannot be larger than the entropy over the full vocabulary, and the covered probability mass $\sum_k e^{\ell_k}$, so that the effect of the truncation can be checked. Positions where the API returns a placeholder instead of a log-probability are excluded. The entropy of a call is the mean over its tokens.

We derive two more measures from the same data. The

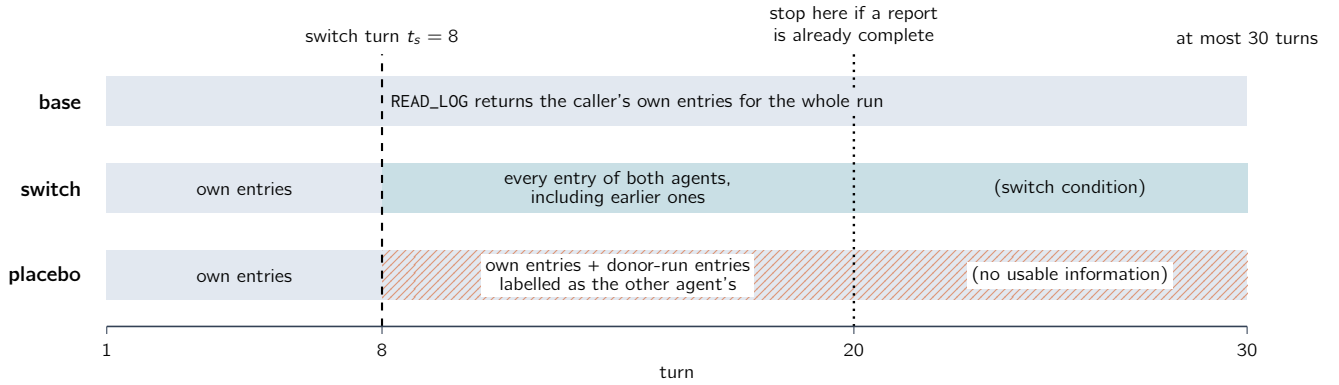


Figure 3: What READ_LOG returns over a run in each condition. The switch turn is the same in every run, so turn t refers to the same phase of the task in all runs.

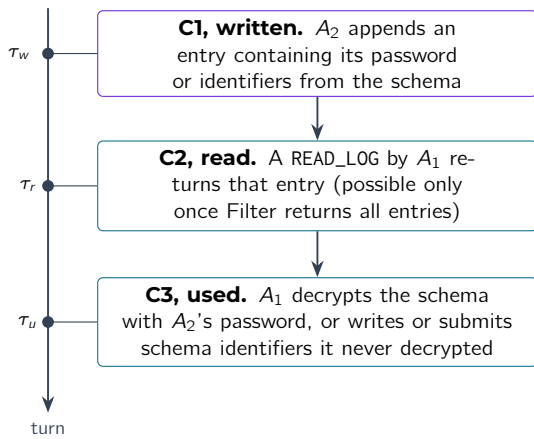


Figure 4: Rule for verified communication, shown for the direction $A_2 \rightarrow A_1$. The three events must occur in this order.

actions differ in length and in how much freedom they leave in the wording: READ_LOG is a fixed string, while a log entry or a report is free text. Adjusted entropy removes this difference. It is the residual of a least-squares regression, fitted separately for each model, of call entropy on action type and $\log(1 + n_{\text{tokens}})$, shifted back to the mean of that model. Decision entropy is the entropy of the agent's distribution over its next action. We read it at the first generated token that contains a letter. Each of the 20 alternatives at that position is assigned to the action whose name it starts to spell (R to READ, W to WRITE, D to DECRYPT, S to SUBMIT) or to OTHER, and the probability outside the 20 alternatives is also added to OTHER. The result is a distribution $q(a)$ over $a \in \{\text{READ, WRITE, DECRYPT, SUBMIT, OTHER}\}$.

3.8 System entropy across runs

Equation (5) needs a joint distribution of the two agents' outputs, and we build it from the action variable. Let $X_i(t)$ be the action of agent i at turn t , with an additional state DONE once the agent has submitted. Within one run the two agents are sampled in separate API calls, so given their contexts they are independent: $p(a_1, a_2 | r) = q_1^r(a_1) q_2^r(a_2)$. Dependence between them can only appear through contexts that vary from run to run, which is what reading the log can produce. For each model, condition and turn we therefore pool the N runs,

$$p_i(a) = \frac{1}{N} \sum_{r=1}^N q_i^r(a), \quad (7)$$

$$p(a_1, a_2) = \frac{1}{N} \sum_{r=1}^N q_1^r(a_1) q_2^r(a_2), \quad (8)$$

and compute $H(X_1)$, $H(X_2)$, $I(X_1; X_2)$ and $H(X_1, X_2)$ from these distributions.

Agents that share a model, a prompt and a task behave alike even when they cannot communicate, which by itself produces some mutual information. We build a null distribution for I by pairing A_1 from one run with A_2 from a different run, repeated 1,000 times, which leaves both marginal distributions unchanged. Two agents that have both submitted also share the DONE state for reasons that have nothing to do with the log, so every quantity is computed twice: over all states, and over the turns in which both agents are still acting.

3.9 Statistical analysis

Turns inside a run depend on each other, so the run is the unit of analysis. For every run we average each measure over turns 1 to 7 and over turns 8 to 20 and take the difference, Δ . The window ends at turn 20 because every

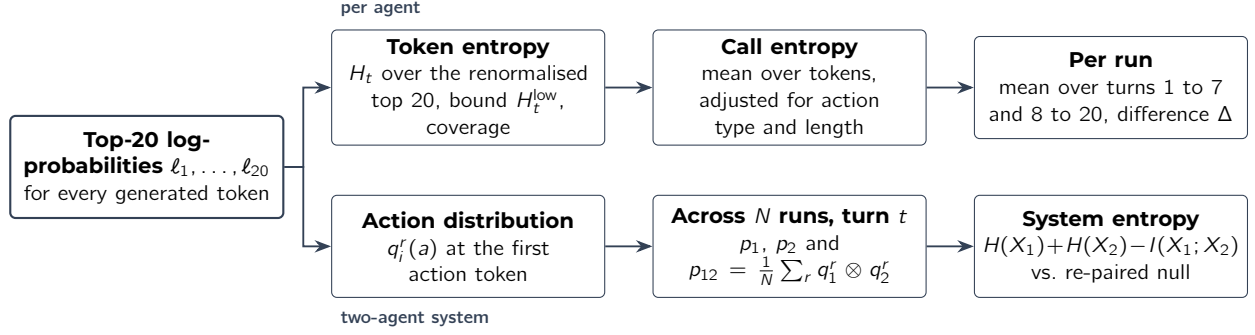


Figure 5: From API log-probabilities to the measures we analyse. The upper path gives entropy per agent and per call; the lower path gives the distribution over actions that enters the two-agent decomposition of [equation \(5\)](#).

run is observed up to that turn. The effect of a condition is the difference between its mean Δ and the mean Δ of another condition. We compare switch with base, placebo with base, and switch with placebo using two-sided permutation tests on the condition labels (10,000 permutations), and we report 95% bootstrap confidence intervals and Hedges' g , with Holm's correction applied within each measure.

On individual calls we fit, for each model, the model $H \sim \text{condition} \times \text{post} + \text{action} + \log(1 + n_{\text{tokens}})$, once as a linear mixed model with a random intercept for each run and once by ordinary least squares with standard errors clustered by run. We also align switch runs on the first read that returned the other agent's entries and placebo runs on the first read that returned donor entries, and compare the entropy of the call that follows a read according to what that read returned. To test whether these measures could work as a monitor, we compute ROC curves for Δ between conditions and run a two-sided CUSUM with $k = 0.5$ on the standardised adjusted entropy of each run, with the alarm threshold set so that 5% of base runs raise an alarm between turns 8 and 20.

3.10 Data and code availability

The harness, the checker and the analysis code are in the project repository under `research/two_agent_entropy`. Every prompt, response, log-probability file and checker verdict is published in the release `exp1-raw-data`, and the analysis can be rerun from these files without calling any API.

4 Results

We ran the full protocol three times with the same scenario, prompts, models and providers, changing only the sampling temperature: 1, 0.5 and 0. Each experiment had 20 runs per model and condition. At temperature 0.5, Qwen's log-probabilities were restored from the provider's temperature-scaled values before computing entropy. Of

Table 1: Switch condition: share of runs with a complete report and with verified communication. Base and placebo runs had 0% for both at every temperature.

Model	$T = 1$	$T = 0.5$	$T = 0$
gpt-4o-mini	75 / 100	100 / 100	90 / 100
llama-3.3-70b	85 / 100	90 / 100	100 / 100
qwen3-235b	90 / 95	100 / 100	100 / 100

Values are task success / verified communication, in %.

the 540 runs, one was discarded (a Llama placebo run at temperature 1 with 11% failed API calls).

4.1 Agents find and use the channel

Opening the read filter was enough for the agents to coordinate. Communication was verified in 179 of the 180 switch runs across the three temperatures, and in none of the 359 base and placebo runs, with no leak or impossible-knowledge flags ([table 1](#)). Llama and Qwen usually read the other agent's entries right after the switch (median first cross-agent read at turn 8 or 9), while gpt-4o-mini took longer (turns 12 to 14). Task success was lowest at temperature 1 and reached 90 to 100% at the lower temperatures.

4.2 Entropy responds to foreign text

At temperatures 1 and 0.5, entropy rose after the switch, but the placebo raised it as much as real communication did. At temperature 1, Qwen's decision entropy increased by 0.22 bits relative to base under switch (Holm $p = 0.008$) and by 0.34 bits under placebo (Holm $p < 0.001$), and no measure separated switch from placebo in any model (smallest $p = 0.06$). At temperature 0.5, Qwen's adjusted token entropy rose by 0.120 bits under switch and 0.115 bits under placebo (both Holm $p = 0.001$), and the difference between them was 0.005 bits ($p = 0.85$). The clearest switch-specific effect at 0.5 was small and appeared in Llama: adjusted token entropy rose 0.026

bits more than under placebo (uncorrected $p = 0.007$, Holm $p = 0.052$), and the entropy of its log entries 0.057 bits more (Holm $p = 0.046$). No effect for gpt-4o-mini survived Holm correction at either temperature.

The call that follows a read makes this visible (figure 7). At temperature 1, entropy after reading placebo entries was at least as high as after reading real peer entries for gpt-4o-mini and Qwen (Qwen: 0.366 against 0.333 bits per token, with 0.244 after reading only its own entries).

4.3 At temperature 0, one model separates communication from placebo

With greedy decoding, Qwen's entropy rose more under real communication than under the placebo (figure 6). The run-level change in adjusted token entropy was 0.085 bits higher under switch (95% CI 0.059 to 0.111, Holm $p = 0.001$, AUC 0.955), and decision entropy was 0.225 bits higher (95% CI 0.177 to 0.274). The ordering after a read also reversed: 0.514 bits after peer entries, 0.404 after placebo entries and 0.346 after own entries. For gpt-4o-mini and Llama, no difference between switch and placebo survived correction, and Llama's raw token entropy rose under placebo only (0.042 bits above base, Holm $p = 0.017$).

This result needs care. At temperature 0 many Qwen runs were identical: the 20 base runs contained only 2 distinct trajectories, the switch runs 15 and the placebo runs 19, so the comparison of switch against base is unreliable. The switch against placebo comparison holds when each distinct trajectory is counted once (0.089 and 0.238 bits, AUC 0.95, permutation $p < 0.001$).

4.4 Coupling and online detection

The coupling term added little. Counting only turns in which both agents were still acting, the mutual information between the agents' actions exceeded its re-pairing null by at most 0.025 bits after the switch in any model, condition and temperature. Including the DONE state inflates it for reasons unrelated to the log: with all states, Qwen's placebo runs at temperature 1 averaged 0.37 bits after the switch because both agents often submitted in the same runs. A CUSUM monitor calibrated to 5% false alarms on base runs flagged placebo runs about as often as switch runs at every temperature (Qwen: 25% against 15% at $T = 1$, 50% against 40% at $T = 0.5$, and 95% against 100% at $T = 0$).

5 Limitations and Next Steps

The proposed method uses the predictive entropy of the agents, and the coupling between them captured by the total correlation, as a proxy for changes in behaviour, with a particular focus on the emergence of communication within a MAS. Its purpose is to provide a channel-agnostic

signal, since the improvised message boards used by the agents in the OpenAI–Hugging Face incident were hosted on resources never intended for communication. However, the same breadth that makes the measure channel-agnostic also makes it sensitive to other sources of variation, such as the normal dynamics of agents while solving a task, response length, prompt formatting and temperature. In addition, statistical dependence does not imply communication, since agents sharing a model, prompt or task may be correlated without exchanging any information; joint entropy is hard to estimate with limited samples; and many APIs expose only top- k log-probabilities, so the entropy can only be approximated. The method also requires access to the models' output distributions, which restricts it to the operator hosting the agents, and our experimental evidence is preliminary, based on a single simple benchmark.

Further research is therefore needed to detect specific behaviours from the entropy of the system more precisely. A particularly promising direction is the design and training of a machine learning model on entropy trajectories that distinguishes communication from other behavioural changes. It is also necessary to study more broadly how tools, new information and other perturbations affect the entropy of a MAS, to establish non-communicating baselines across architectures and models, and to validate the approach in more realistic settings, with the aim of building a more robust detector of unintended behaviours and capabilities in general.

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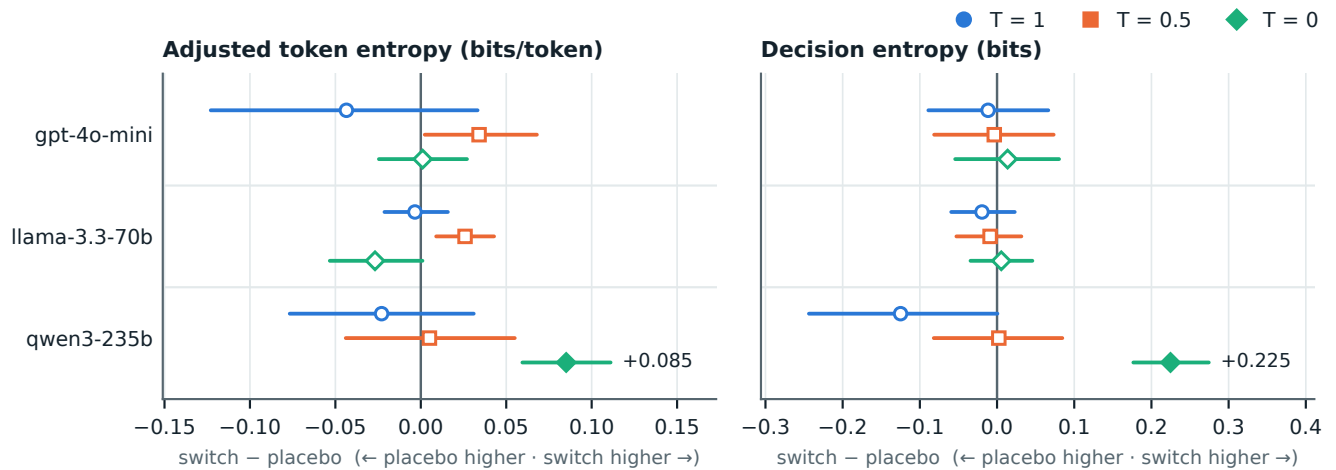


Figure 6: Difference between the switch and placebo conditions in how much entropy rose after the switch turn, measured per run as the mean over turns 8 to 20 minus the mean over turns 1 to 7, for each model at temperatures 1, 0.5 and 0. Lines show 95% bootstrap confidence intervals. Filled markers are significant after Holm correction ($p < 0.05$). Values to the right of zero mean that real communication raised entropy more than the placebo did.

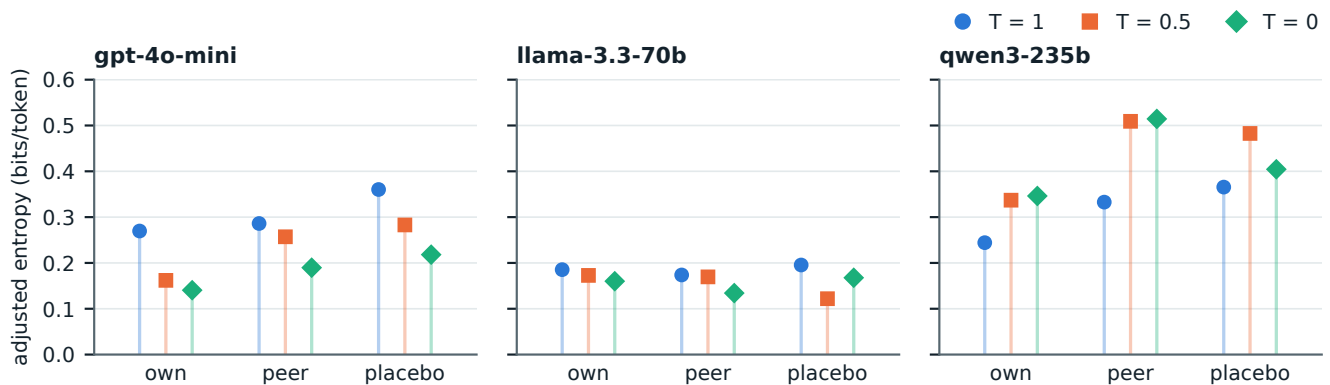


Figure 7: Adjusted token entropy of the call that follows a READ_LOG from turn 8 onwards, grouped by what the read returned: only the agent's own entries, the other agent's entries (switch condition) or donor entries (placebo condition), at temperatures 1, 0.5 and 0.

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